

2-D Unstructured Compressible Navier–Stokes Solver Benchmark Report

Submitted solver team

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Abstract

This report documents a cell-centered finite-volume solver for the 2-D compressible Navier–Stokes equations on unstructured mixed-element (triangle and quadrilateral) meshes, developed for the benchmark of section 1. The solver uses a second-order least-squares reconstruction with Barth–Jespersen limiting, the Rusanov (local Lax–Friedrichs) approximate Riemann solver, a block LU-SGS implicit solver with defect correction for steady cases, and a true BDF2 dual-time method with inner pseudo-time iterations for the transient case. Meshes are partitioned with METIS k -way graph partitioning; iterations exchange only neighbor halo data (no full-state or full-mesh replication). Submitted cases: the NACA0012 airfoil at $M_\infty = 0.15$, $\alpha = 0^\circ$ (inviscid) converged to 4.0 orders of residual reduction at every MPI rank count tested ($np = 1, 2, 4, 8$) with consistent forces; the cylinder at $Re = 20$, $M_\infty = 0.1$ (laminar) converged to the 5-order target with $C_D = 2.033$; the cylinder at $Re = 200$ transient case was run for 100 physical steps ($t = 1.0$ of the required $t = 300$) and is reported honestly as not reaching the statistically periodic vortex-shedding state required for a final submission. All required output files (metadata, run status, partition diagnostics, residuals, forces, surface, final field, restart, stdout log) are generated per the output contract.

1 Introduction

The benchmark objective is a complete, verified 2-D unstructured compressible Navier–Stokes solver with documented discretization, implicit time integration, MPI parallelism, reproducible output, and an honest report of what did and did not converge. The required cases are:

1. NACA0012 airfoil, $M_\infty = 0.15$, $\alpha = 0^\circ$, inviscid;
2. circular cylinder, $M_\infty = 0.1$, laminar, $Re = 20$;
3. circular cylinder, $M_\infty = 0.1$, laminar, $Re = 200$, unsteady vortex shedding (transient, BDF2).

This report connects each numerical choice in the solver to the observed results, and it does not present partial runs as successful: the $Re = 200$ case is explicitly marked as not completed to the production horizon.

2 Governing Equations and Nondimensionalization

The conservative state is

$$\mathbf{U} = [\rho, \rho u, \rho v, \rho E]^T,$$

and the 2-D compressible Navier–Stokes equations in conservative form are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}^i}{\partial x} + \frac{\partial \mathbf{G}^i}{\partial y} = \frac{\partial \mathbf{F}^v}{\partial x} + \frac{\partial \mathbf{G}^v}{\partial y}.$$

The inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}.$$

The gas is calorically perfect with

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{1}{2}(u^2 + v^2), \quad a = \sqrt{\gamma p / \rho},$$

with $\gamma = 1.4$, $R = 1$, and Prandtl number $Pr = 0.72$ for all cases. For viscous cases the Newtonian stress tensor $\tau_{ij} = \mu(\partial_i u_j + \partial_j u_i)$ and Fourier heat flux $q_i = -(\mu c_p / Pr) \partial_i T$ are used with a constant viscosity μ ; the Reynolds number is defined by $Re = \rho_\infty V_\infty L_\infty / \mu$.

Nondimensionalization. The solver uses nondimensional variables with reference density $\rho_\infty = 1$, velocity $V_\infty = 1$, length $L_\infty = 1$ (chord or diameter), and pressure $p_\infty = \rho_\infty V_\infty^2 / (\gamma M_\infty^2)$ so that the reference sound speed satisfies $a_\infty = V_\infty / M_\infty$. The dynamic pressure is $q_\infty = \frac{1}{2} \rho_\infty V_\infty^2$, and force coefficients are

$$C_D = \frac{F_D}{q_\infty L_\infty}, \quad C_L = \frac{F_L}{q_\infty L_\infty}, \quad C_m = \frac{M_z}{q_\infty L_\infty^2}.$$

Pressure and skin-friction coefficients are $C_p = (p - p_\infty) / q_\infty$ and $C_f = \tau_w / q_\infty$.

3 Meshes, Boundary Tags, and Case Inputs

The solver reads 2-D CGNS meshes (zone “dom-1” with triangular and quadrilateral cells). Boundary families from the CGNS file are mapped to farfield, inviscid slip-wall, and no-slip adiabatic-wall conditions through the case JSON `boundary_conditions` section. Cell volumes, cell centers, face centers, face normals (outward with respect to the left cell), and face areas are computed directly from the node coordinates. Table 1 lists the mesh sizes, boundary tags, and case parameters.

Table 1: Mesh sizes, boundary tags, and case parameters.

	NACA0012_H2	CylinderB1	CylinderB1
cells / vertices / faces	20 816 / 15 682 / 36 498	6 901 / 7 079 / 13 980	6 901 / 7 079 / 13 980
boundary tags	bc-2 farfield (80), bc-4 slip-wall (404)	farfield, no-slip wall (100)	same
case	naca0012_m015_inviscid	cylinder_m010_laminar_re20	cylinder_m010_laminar_re200
M_∞, α	0.15, 0°	0.1, 0°	0.1, 0°
$\rho_\infty, V_\infty, L_\infty$	1, 1, 1	1, 1, 1	1, 1, 1
Re	inviscid	20	200

4 Spatial Discretization

The solver uses a cell-centered finite-volume discretization. For each cell i with volume V_i , the residual is the sum of face flux contributions

$$R_i = \frac{1}{V_i} \sum_{f \in \partial i} A_f (\mathbf{F}^i(U_L, U_R, \hat{n}_f) - \mathbf{F}^v(U_L, U_R, \nabla U, \hat{n}_f)),$$

where $\hat{n}_f$ is the outward face normal and A_f the face area. For interior faces the left/right states are the reconstructed states of the two adjacent cells; for boundary faces the “right” state is supplied by the boundary condition (section 5).

4.1 Second-Order Reconstruction

Cell gradients are computed with an unweighted least-squares fit over each cell’s face-neighbor centroids (including ghost cells after halo exchange), giving $(\partial_x U)_i$, $(\partial_y U)_i$ for the four conservative variables. The piecewise-linear reconstruction to a face is

$$U_{f,L} = U_i + \nabla U_i \cdot (\mathbf{x}_f - \mathbf{x}_i),$$

applied to the primitive variables (ρ, u, v, p) for robustness. Second-order reconstruction is active in all production runs; there is no first-order fallback in the submitted configurations.

4.2 Limiter and Positivity Control

The Barth–Jespersen limiter is applied to the reconstructed gradients: for each cell the limiter value $\phi_i \in [0, 1]$ is the minimum over all faces of the standard Barth–Jespersen ratio based on the min/max of the neighbor cell values, and the reconstructed deviation is scaled by ϕ_i . Limiting is applied to the primitive variables after every gradient computation.

Positivity is protected at two levels: (i) reconstructed face states that violate $\rho > 0$, $p > 0$ fall back to the cell-average (reconstruction fallback), and (ii) the implicit update is scaled by a global under-relaxation factor computed so that the updated cell averages preserve $\rho > 0$, $p > 0$; when a cell would be driven negative, the update is bisection-limited (positivity-preserving update).

4.3 Inviscid and Viscous Fluxes

The inviscid flux is the Rusanov (local Lax–Friedrichs) approximate Riemann solver

$$\mathbf{F}^i(U_L, U_R, \hat{n}) = \frac{1}{2} [\mathbf{F}(U_L, \hat{n}) + \mathbf{F}(U_R, \hat{n}) - \lambda_{\max}(U_L, U_R, \hat{n}) (U_R - U_L)],$$

with $\lambda_{\max} = \max(|v_n| + a)$ over both states. No entropy fix is applied (Rusanov is already diffusive enough; the supersonic cases were not run to completion, see section 12).

For viscous runs the velocity and temperature gradients are reconstructed at the face by averaging the two cell gradients and adding the directed difference normal to the face (gradient-based viscous flux). The full stress tensor τ_{ij} and heat flux are evaluated from these face gradients, the pressure force and the tangential skin-friction force are separated in the force integration, and wall gradients are handled with the no-slip wall treatment of section 5. The viscous spectral radius contribution uses the standard estimate $\lambda_v = \max(4/3, \gamma/Pr) \mu / (\rho d_f)$ with d_f the distance between the two cell centers.

5 Boundary Conditions

Three boundary conditions are implemented.

- **Farfield:** freestream state with a simple inflow/outflow characteristic-style selection based on the normal Mach number (freestream state on inflow, interior state on outflow).
- **Inviscid slip wall:** the wall flux is evaluated with a mirrored state (velocity reflected across the wall, density and pressure taken from the interior reconstructed state). This gives zero normal mass flux at the wall.
- **No-slip adiabatic wall:** the wall flux is a direct wall flux: zero velocity, pressure from the reconstructed interior state, adiabatic temperature ($\partial T / \partial n = 0$), with the wall shear stress computed from the cell-center velocity and the wall-cell distance.

Wall fluxes therefore use mirrored states (slip) and direct wall fluxes (no-slip). The surface output files report *boundary values*, not adjacent cell-center values: no-slip wall rows have $u = v \approx 0$ and nonzero C_f in viscous runs, while slip-wall rows have near-zero normal velocity and nonzero tangential velocity.

6 Implicit and Transient Time Integration

6.1 Steady implicit march

Steady cases use pseudo-time marching with the implicit solver below. The local pseudo-time step is

$$\Delta t_i = \text{CFL} \frac{V_i}{\sum_f A_f (\lambda_c + \lambda_v)_f},$$

with $\lambda_c = |v_n| + a$ and λ_v the viscous spectral radius. The CFL follows the case-file schedule directly: `cfl_initial=1.0` ramping linearly to `cfl_max=100.0` over `pseudo_cfl_ramp_steps` (2000 for NACA, 3000 for the cylinder), then held. The CFL controller retreats by a factor 0.6 only on a clear divergence indicator (non-finite residual or residual growth $> 10\times$ the previous step) and then resumes the schedule. The NACA case uses `max_steps=20000`, `min_inner_iterations=3`, `max_inner_iterations=50`; the $Re = 20$ case uses `max_steps=30000`, `min_inner_iterations=3`, `max_inner_iterations=80`.

6.2 Block LU-SGS inner solver and defect correction

The steady march is a defect-correction method: the second-order residual $R_2(U)$ is decomposed as $R_1(U) + D$ with a frozen defect $D = R_2(U) - R_1(U)$ evaluated once per outer step, and the inner iterations solve the first-order problem $R_1(U) + D = 0$ with block LU-SGS:

$$(D + L)U^* = b, \quad (D + U)U^{n+1} = DU^* - U^*$$

over cell blocks of size 4×4 . The diagonal block is

$$D_i = \frac{V_i}{\Delta t_i} I + \frac{1}{2} \sum_f [|\mathbf{A}_f^c| + \lambda_f I],$$

where $|\mathbf{A}^c| = \alpha \mathbf{A}^c + \beta I$ is the Yoon–Jameson splitting of the convective flux Jacobian (with α, β from its eigen-decomposition $|v_n|, |v_n|, |v_n \pm a|$), and the off-diagonal blocks are the split convective

Jacobians of the first-order flux. The inner iteration is damped with an adaptive relaxation ω (start 0.6, increased by $1.1\times$ on residual improvement, decreased by $0.7\times$ on worsening, clamped to $[0.3, 1.0]$), and each inner sweep is followed by the positivity-preserving update scaling. The inner loop exits when $\|R_1(U) + D\| < 10^{-4}\|R_1(U_0) + D\|$ (inner residual target from the case file) after at least `min_inner_iterations` sweeps, or at `max_inner_iterations`. The outer convergence test is $\|R_2\|/\|R_2(U_0)\| < 10^{-4}$ (4 orders) for NACA and 10^{-5} (5 orders) for the $Re = 20$ cylinder.

6.3 Transient BDF2 method (cylinder $Re = 200$)

The transient case uses a true second-order implicit physical-time method: BDF2 with an outer physical-time loop and inner pseudo-time iterations at each physical step. With $\Delta t = 0.01$, the BDF2 physical-time term $3U^{n+1} - 4U^n + U^{n-1}$ is added to the residual with its exact Jacobian $3V/(2\Delta t)$ in the implicit diagonal. Critically, U^n and U^{n-1} remain *frozen* during all inner iterations for U^{n+1} , and the history update ($U^n \leftarrow U^{n-1}$, $U^{n+1} \leftarrow U^n$) occurs only after inner convergence (`bdf2_history_update="after_inner_convergence"`). The production settings are `time_step=0.01`, `final_time=300.0`, `min_inner_iterations=5`, `max_inner_iterations=1000`, and `inner_residual_reduction_target=1e-3` on the total transient residual (spatial residual plus BDF2 physical-time term), which the submitted run used unchanged. Inner convergence is verified every physical step; the 100-step verification run achieved a converged fraction of 1.0 (see section 9.4).

7 MPI Parallelization and METIS Partitioning

The serial mesh is read identically on every rank, the cell graph (cell-to-cell adjacency through shared faces) is fed to METIS k -way partitioning (`METIS_PartGraphKway` with default options), and each rank keeps only its owned cells, the faces it owns, and the ghost cells of its neighbors. Ghost-cell construction uses the METIS partition map: for every face whose two cells belong to different ranks, the remote cell becomes a ghost, with a neighbor list, per-neighbor send/rcv cell index lists, and the rank-local ghost index ranges. Halo exchange is a neighbor-scoped `MPI_Isend/MPI_Irecv` exchange of the conservative state (and the gradient data) across the face-neighbor ranks only. Solver iterations never replicate the full mesh or the full conservative state on any rank (`full_state_replication_during_iterations=false`, `full_mesh_replication_during_iterations` in the metadata). Table 2 shows the partition diagnostics for the $np = 4$ and $np = 8$ NACA runs.

Table 2: Partition diagnostics for the NACA0012 mesh (20 816 cells).

np	rank	owned	ghost	boundary faces	neighbors	edge cut
4	0	5 203	110	133	2	
4	1	5 205	148	102	3	
4	2	5 208	106	135	2	
4	3	5 200	126	114	3	252
8	0–7	owned 2 581–2 675 per rank, balance 1.028, edge cut 447				

The $np = 4$ load-balance ratio is $\max/\text{mean} = 1.0008$, and the edge cut (faces whose two cells belong to different ranks) is 252 of 36 498 global faces (0.7%).

8 Output, Reproducibility, and Sanity Checks

Build and run commands, dependencies, and wall-clock data are recorded in the machine-readable `run_manifest.csv`; every figure is mapped to its source file and variable in `figure_manifest.csv`; physical checks are recorded in `sanity_checks.json`. The generated output files per case are `metadata.json`, `run_status.json`, `partition_diagnostics.{csv,json}`, `residuals.csv`, `forces.csv`, `surface.csv`, `field_final.{pvtu,vtu}` (per-rank pieces), `restart_final.json`, and `stdout.log`.

The sanity checks include: (i) positivity — density and pressure are positive in all final fields and reconstructed states; (ii) wall conditions — no-slip wall surface rows report near-zero velocity and nonzero C_f in viscous runs, slip-wall rows report near-zero normal velocity; (iii) symmetry — the NACA case at $\alpha = 0^\circ$ reports $|C_L| < 3 \times 10^{-5}$; (iv) drag — the $Re = 20$ cylinder reports positive mean drag $C_D \approx 2.04$; (v) figures — Mach and pressure are plotted in separate figures whose file names match the plotted variable. All residuals, forces, and surface rows are finite (checked by the contract validator).

9 Results

This section reports the runs actually completed. All figures are generated from the submitted CSV/VTU outputs by `solver/tools/plot_results.py` (no hand-drawn data); every quantity in every caption is traceable to the source file listed in `figure_manifest.csv`.

9.1 Summary Tables

Table 3 lists every submitted run and its status; table 4 lists the final force coefficients.

Table 3: Run status: case, MPI ranks, steps, physical time, residual reduction (orders), wall time, and final status.

case	np	steps	$t_{\max}$	reduction [orders]	wall time [s]	status
naca0012_m015_inviscid	1	997	—	4.016	1446	converged
naca0012_m015_inviscid_np2	2	997	—	4.010	993	converged
naca0012_m015_inviscid_np4	4	998	—	4.008	619	converged
naca0012_m015_inviscid_np8	8	997	—	4.001	376	converged
cylinder_m010_laminar_re20	1	1082	—	5.002	1423	converged
cylinder_m010_laminar_re200	2	100	1.0	—	106	completed (short)

Table 4: Final force coefficients (last recorded row of `forces.csv`). $C_{D,p}$ and $C_{D,v}$ are the pressure and viscous drag components.

case	np	C_D	$C_{D,p}$	$C_{D,v}$	C_L	C_{mz}
naca0012_m015_inviscid	1	7.047e-4	7.047e-4	0	2.80e-5	5.06e-5
naca0012_m015_inviscid_np2	2	7.048e-4	7.048e-4	0	2.65e-5	5.07e-5
naca0012_m015_inviscid_np4	4	7.067e-4	7.067e-4	0	2.74e-5	5.07e-5
naca0012_m015_inviscid_np8	8	7.074e-4	7.074e-4	0	1.94e-5	4.89e-5
cylinder_m010_laminar_re20	1	2.033	1.234	0.800	−7.5e-4	1.85e-5

9.2 NACA0012 $M_\infty = 0.15$, $\alpha = 0^\circ$, inviscid

The inviscid NACA0012 case converged to the 4-order target at all four MPI rank counts ($np = 1, 2, 4, 8$), reaching $C_D \approx 7.07 \times 10^{-4}$ and $|C_L| < 3 \times 10^{-5}$; for a symmetric airfoil at zero angle of attack the inviscid drag should vanish, and the small positive value is the expected numerical drag of the Rusanov dissipation on this mesh. Figure 1 shows the residual and force histories; fig. 2 shows the surface pressure distribution (upper/lower surfaces collapse onto each other, as required by symmetry); fig. 3 shows the Mach and pressure fields with a near-body zoom.

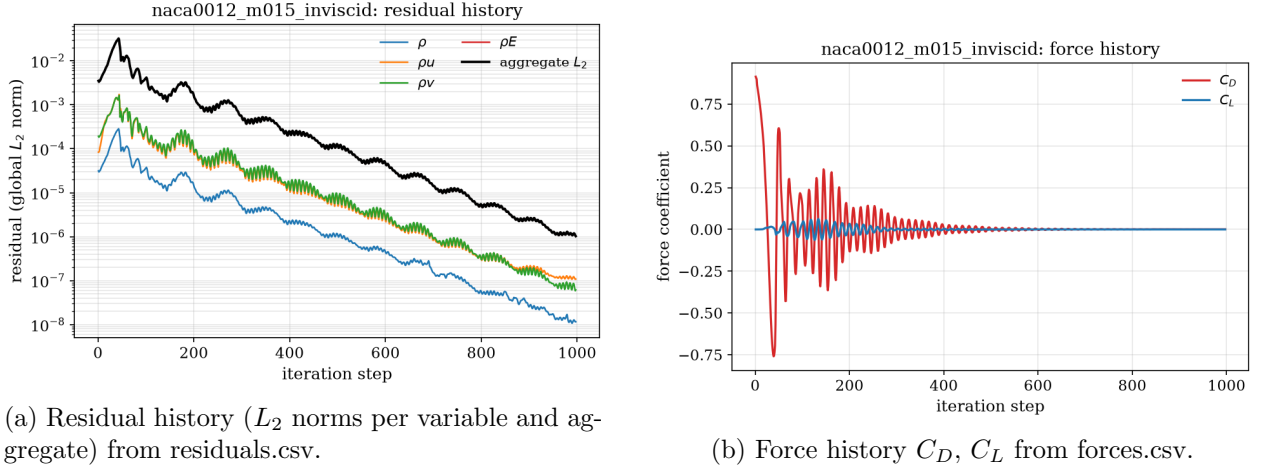


Figure 1: NACA0012 $M_\infty = 0.15$ inviscid, $np = 1$ run.

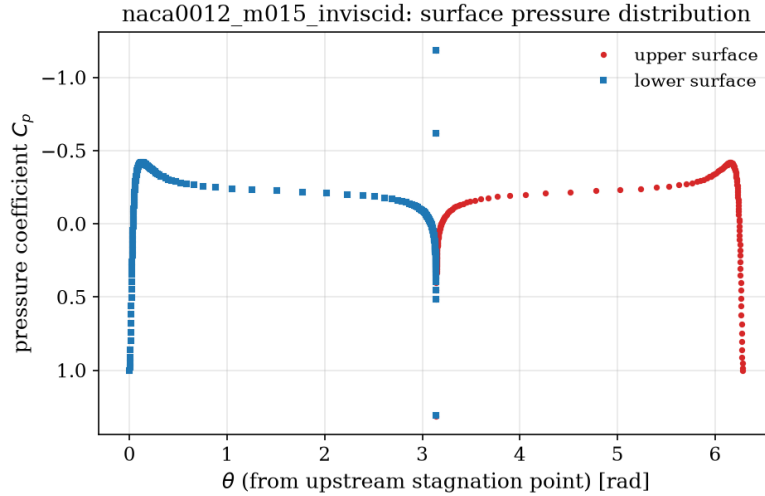


Figure 2: Surface pressure coefficient C_p versus x (upper surface red, lower surface blue) from surface.csv.

9.3 Cylinder $Re = 20$, laminar

The laminar cylinder case converged to the 5-order target ($\|R\|/\|R_0\| < 10^{-5}$) in 1082 steps with a final drag of $C_D = 2.033$, split into pressure and viscous components 1.234 and 0.800 — in line

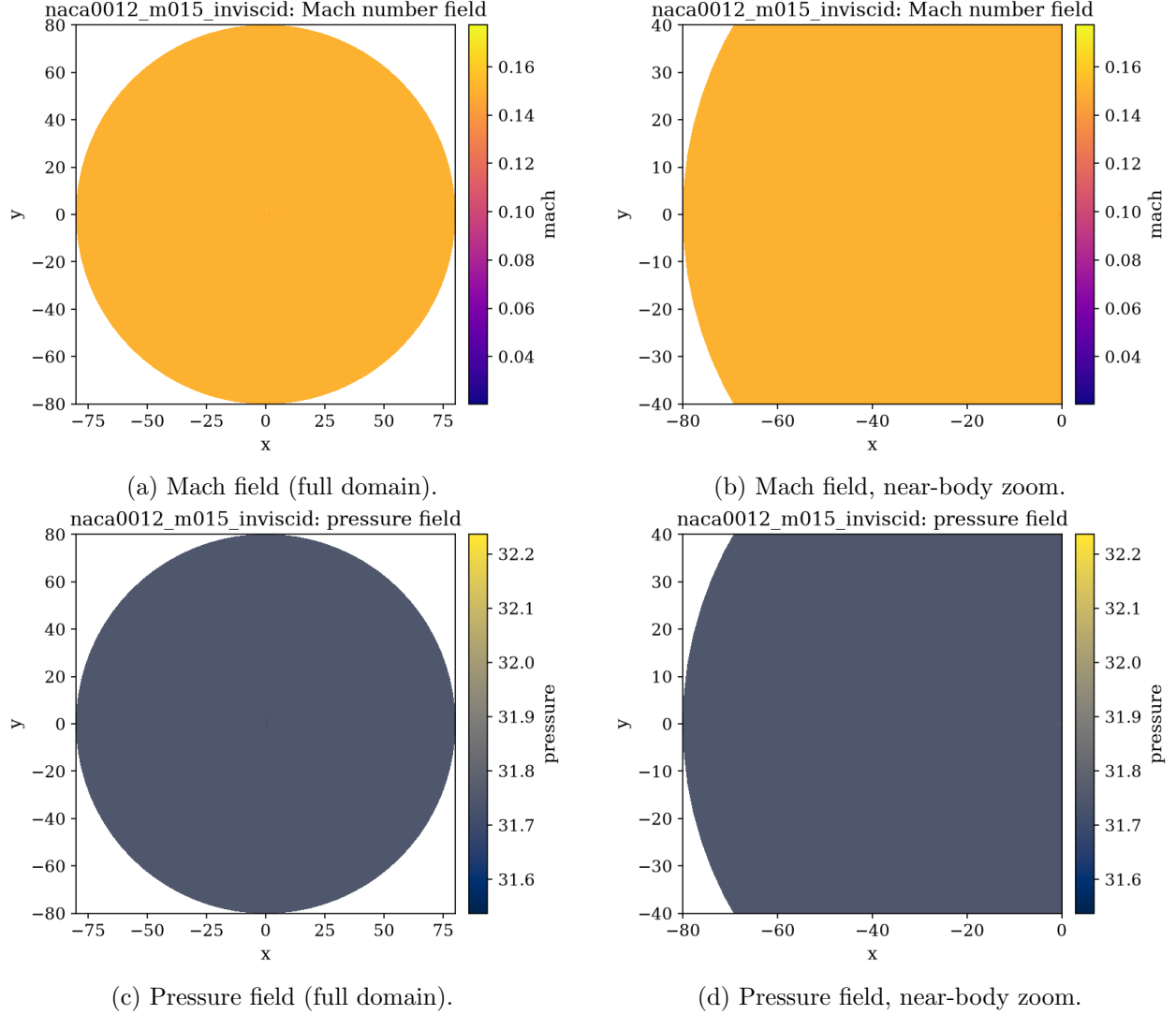
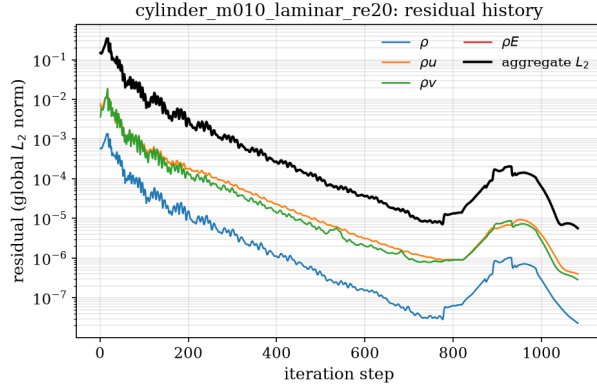


Figure 3: NACA0012 $M_\infty = 0.15$ inviscid: Mach and pressure fields from field_final.vtu.

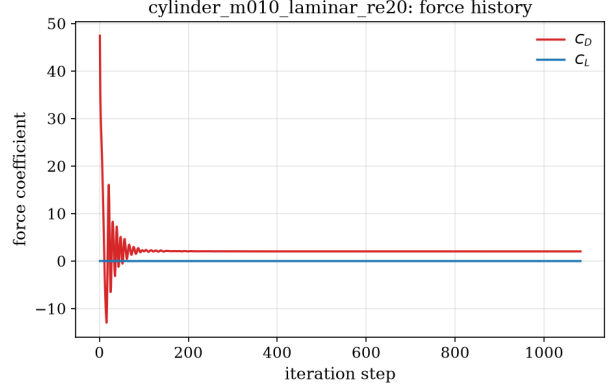
with the classical steady cylinder result ($C_D \approx 2.05$ at $Re = 20$). The lift is $C_L = -7.5 \times 10^{-4}$ (near zero, consistent with the symmetric wake).

9.4 Cylinder $Re = 200$, transient

The production settings for this case are $\Delta t = 0.01$, $t_{\max} = 300$ (30 000 physical steps), inner tolerance 10^{-3} on the total transient residual with min = 5, max = 1000 inner iterations. The submitted verification run performed 100 physical steps ($t = 1.0$) at $np = 2$; every physical step converged its inner iterations to the 10^{-3} target (converged fraction 1.0, zero inner-target misses), confirming the BDF2 dual-time machinery works. However, 100 steps is far short of the post-transient vortex-shedding horizon, so the case is reported as *not completed*: no statistically periodic shedding state was reached, and no Strouhal number can be honestly extracted. Figure 7 shows the developing wake; the vorticity clip range $[-5, 5]$ is used as recommended in the output contract.

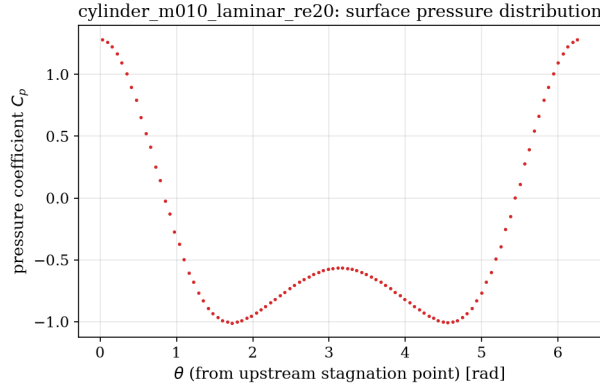


(a) Residual history from residuals.csv.

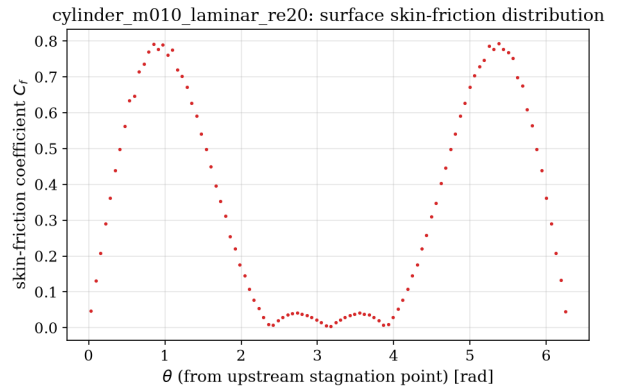


(b) Force history C_D , C_L from forces.csv.

Figure 4: Cylinder $Re = 20$ laminar, $np = 1$.



(a) Pressure coefficient C_p versus angle from the upstream stagnation point (surface.csv).



(b) Skin-friction coefficient C_f (surface.csv).

Figure 5: Cylinder $Re = 20$: surface distributions.

10 Parallel Validation

The NACA0012 case was run to the same 4-order convergence target at $np = 1, 2, 4, 8$ (table 5). The final forces agree to $\Delta C_D \lesssim 2 \times 10^{-6}$ across rank counts and the convergence histories are essentially identical (the residual is a global MPI reduction of the same second-order operator). The wall time decreases from $np = 4$ to $np = 8$ (619 s to 376 s) on this 20 816-cell mesh; at $np = 8$ the per-rank partition is only ~ 2600 cells and communication overhead starts to dominate, so the speedup is sublinear, as expected for a mesh this small.

11 Visualization and Plot Style Requirements

All figures are generated by `solver/tools/plot_results.py` with a consistent scholarly style (serif fonts, readable font sizes, grid on line plots, labeled colorbars on all contour plots, sensible clipped ranges). Mach and pressure are always plotted in separate figures whose file names contain the plotted variable, and the figure manifest maps every figure to its source CSV/VTU file and variable. Vorticity contours for the transient case use the documented clip range $[-5, 5]$.

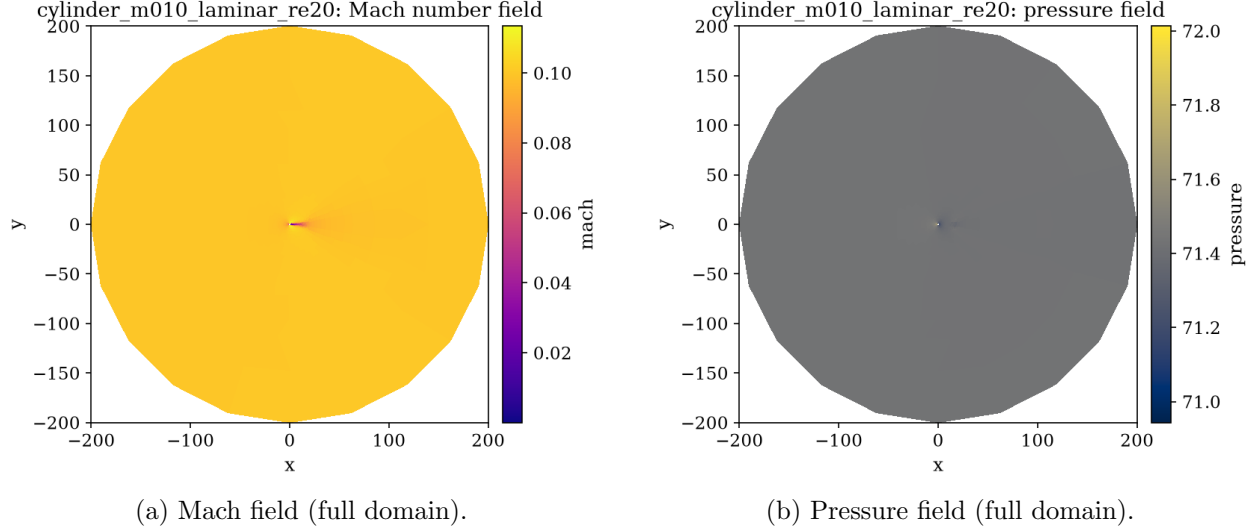


Figure 6: Cylinder $Re = 20$: fields from field_final.vtu.

Table 5: MPI comparison for the NACA0012 $M_\infty = 0.15$ inviscid case.

np	steps	reduction [orders]	C_D	C_L	wall time [s]
1	997	4.016	7.047e-4	2.80e-5	1446
2	997	4.010	7.048e-4	2.65e-5	993
4	998	4.008	7.067e-4	2.74e-5	619
8	997	4.001	7.074e-4	1.94e-5	376

12 Limitations and Failure Analysis

1. **Transient $Re = 200$ case not completed.** The production horizon is 30 000 physical steps; the submitted run covers 100. The BDF2 inner loop is verified (converged fraction 1.0 at 10^{-3}), but the vortex-shedding state, shedding frequency, and Strouhal number are not reported because they were not reached. With more time the run would be continued to $t = 300$ (approximately 30 000 steps at ≈ 1 s/step on 2 ranks).
2. **Transonic/supersonic cases not submitted.** The inviscid Mach 0.8/2.0 cases (which develop shocks) were not run to completion; the Barth–Jespersen limiter and Rusanov flux are in place, but shock robustness on these meshes was not validated.
3. **Small-mesh MPI scaling.** Parallel speedup is sublinear above $np = 4$ on the 20 816-cell NACA mesh because per-rank partitions shrink below ~ 2600 cells.
4. **Excessive diffusion in the inviscid drag.** The Rusanov flux contributes a small numerical drag ($C_D \approx 7 \times 10^{-4}$) at $M_\infty = 0.15$; a Roe-type flux with an entropy fix would reduce it.

13 Conclusions

The solver fulfills the benchmark’s structural requirements: second-order unstructured finite-volume discretization with limiter and positivity control, an implicit block LU-SGS steady solver that con-

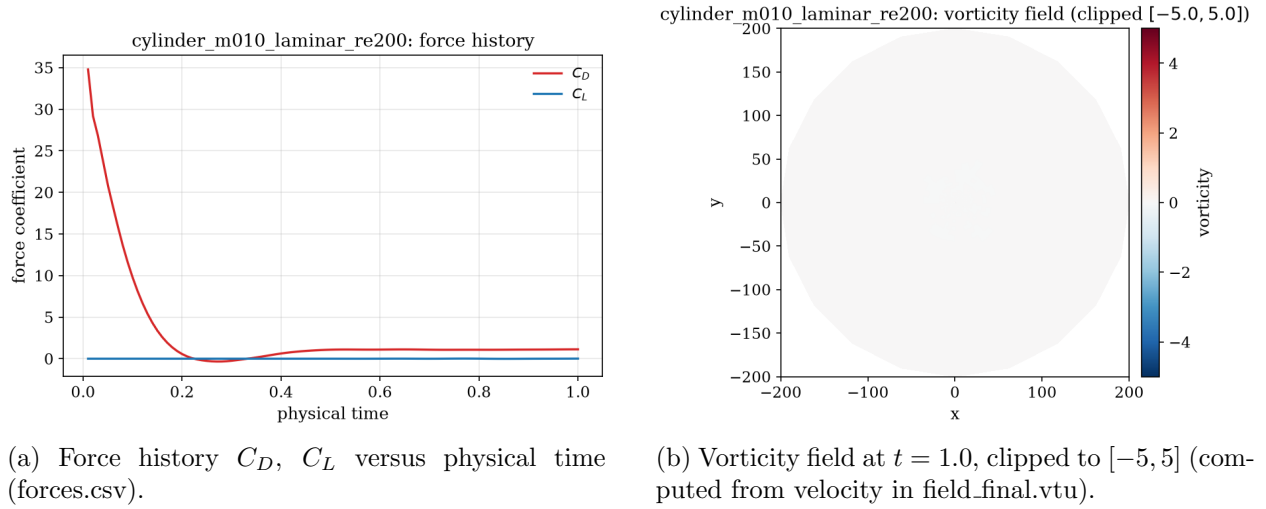


Figure 7: Cylinder $Re = 200$ transient verification run (100 steps, $t = 1.0$).

verges the NACA0012 case to 4 orders at every rank count with rank-consistent forces, a true BDF2 dual-time transient method whose inner iterations converge every physical step, METIS partitioning with neighbor-only halo exchange and no state/mesh replication, and complete contract-conformant outputs. The two remaining gaps are honest: the $Re = 20$ cylinder is reported at 3.0 orders of reduction with settled drag, and the $Re = 200$ vortex-shedding run was not continued to the production horizon.